

$$= 0.5 \%$$

2

D

The goods train will overtake the express train after time t when their displacement from the station is the same.

For the good train: $s_1 = ut = 10t$

For the express train: $s_2 = ut + \frac{1}{2} at^2$
 $= 0 + \frac{1}{2} (0.50)t^2$

When $s_1 = s_2$,

$$10t = \frac{1}{2} (0.50)t^2$$

$$t = 40 \text{ s}$$

3

B

Backward push by back wheel on the road